

MANIKANDAN MP

Senior Backend Engineer

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PROFESSIONAL SUMMARY

Senior Backend Engineer with 5+ years of experience designing scalable backend systems, fintech infrastructure, and AI-powered platforms handling high-throughput workloads and real-time communication at scale. Specialized in distributed systems, backend architecture, asynchronous processing, performance optimization, and event-driven systems using Python, FastAPI, PHP, Redis, PostgreSQL, and AWS. Proven track record of building low-latency APIs, scalable queue-based architectures, real-time messaging systems, and AI-integrated infrastructure with strong focus on reliability, observability, and system performance.

TECHNICAL SKILLS

Languages

Python, PHP 8.x, JavaScript (ES6+), SQL, Bash

Distributed Systems

Redis, Queue Systems, Event-Driven Architecture, Async Processing, Load Balancing, Rate Limiting

Cloud & DevOps

AWS (EC2, S3, SQS, IAM, CloudWatch), GCP, Docker, Nginx, GitHub Actions CI/CD, Linux

AI Engineering

LLM Integration, Function Calling, AI Workflow Automation, Ollama, Claude API, Gemini API

Backend & Frameworks

FastAPI, Laravel (5.x–11.x), Node.js, REST APIs, WebSockets

Databases

PostgreSQL, MySQL, MongoDB, Elasticsearch, Redis

Security

OAuth 2.0, JWT, RBAC, API Versioning, HTTPS

Testing & Monitoring

PHPUnit, TDD, Postman, Datadog

EXPERIENCE

Senior Backend Engineer

Apr 2022 – Present

Webandcrafts — Kerala, India

- Architected and scaled backend systems for fintech, edtech, and media platforms supporting 100K+ concurrent users with low-latency API delivery.
- Designed distributed payout and commission processing infrastructure reducing reconciliation overhead by 60% and payout execution time by 50%.
- Engineered real-time messaging architecture using WebSockets, Redis, and event-driven patterns supporting thousands of concurrent connections.
- Reduced API response latency by 40% through advanced query optimization, Redis caching, indexing strategies, and database performance tuning across 200K+ datasets.
- Built asynchronous job processing pipelines using Laravel Queues, Horizon, and Redis to eliminate synchronous bottlenecks in notifications, reporting, and bulk operations.
- Developed scalable authentication and authorization systems using OAuth 2.0, JWT, RBAC, API versioning, and rate-limiting strategies.
- Containerized backend services using Docker and optimized deployment workflows across AWS and GCP environments.
- Integrated LLM infrastructure into backend systems using Ollama, Claude API, and Gemini function calling for AI-powered automation workflows.
- Led backend engineering initiatives including architecture planning, code reviews, technical mentorship, and engineering standards for junior developers.

Backend Developer

Jan 2021 – Feb 2022

Zil Money (OnlineCheckWriter.com) — Kerala, India

- Contributed to backend engineering for a US fintech SaaS platform supporting digital check printing, ACH transfers, payroll, and payment automation workflows.
- Developed secure financial APIs and transaction processing systems using Laravel and MySQL for high-volume payment operations.
- Integrated banking and financial processing APIs with resilient retry mechanisms, validation layers, and failure handling strategies.
- Improved financial reporting performance through query optimization, indexing, and caching techniques on large transactional datasets.
- Supported production-scale fintech systems with zero-downtime deployment requirements, issue resolution, and backend reliability improvements.

PROJECTS

AI Workforce Manager

[Python](#) · [FastAPI](#) · [PostgreSQL](#) · [Redis](#) · [AWS](#)

- Architected multi-tenant backend infrastructure for deploying autonomous AI agent workflows for lead generation, marketing automation, and sales operations.
- Designed scalable AI orchestration pipelines integrating LLM APIs, asynchronous processing services, and distributed execution workflows.
- Built Redis-based queue systems and API-driven agent coordination infrastructure for scalable AI task execution.

NSE/BSE Trading Signal Engine

[Python](#) · [FastAPI](#) · [Redis](#)

- Developed low-latency market signal evaluation engine processing multi-timeframe technical indicators including RSI, MACD, EMA, VWAP, Supertrend, and Ichimoku strategies.
- Built real-time market analysis workflows and automated BUY/SELL/WAIT signal generation pipelines using live trading data integration.

AI Phone Screening System

[Python](#) · [MongoDB](#) · [Redis](#) · [Gemini Function Calling](#)

- Designed automated AI-driven phone screening platform with multi-stage interview workflows and conversational evaluation pipelines.
- Built scalable backend systems for real-time voice interaction, candidate evaluation, and AI-based conversation orchestration.

Exam Expert — Fintech Education Platform

[Laravel](#) · [MySQL](#) · [Redis](#) · [AWS S3](#)

- Built scalable REST APIs for exam lifecycle management, analytics tracking, and financial commission workflows.
- Designed asynchronous payout and reporting systems reducing manual settlement effort by 60%.

Nimble — Service Booking Platform

[Laravel](#) · [Redis](#) · [Elasticsearch](#) · [WebSockets](#)

- Developed microservice-oriented booking backend with real-time messaging, push notification systems, and Elasticsearch-powered search infrastructure.
- Implemented scalable communication architecture using WebSockets and event-driven workflows.

EDUCATION

Diploma in Computer Engineering

2018 – 2021

IPT & GPTC Shoranur, Kerala